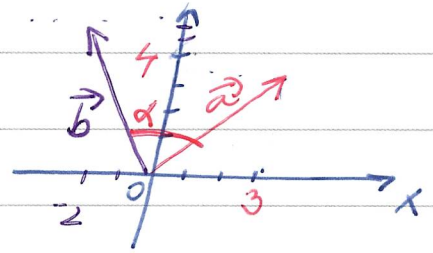


ALGORITMY

PR 7

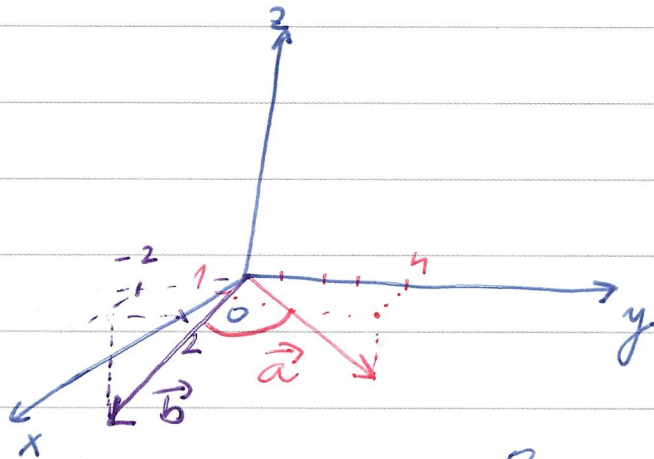
a) $\vec{a} (3, 4)$; $\vec{b} (-2, 6)$



$$\cos(\vec{a}, \vec{b}) = \frac{3 \cdot (-2) + 4 \cdot 6}{\sqrt{3^2 + 4^2} \cdot \sqrt{(-2)^2 + 6^2}} = \frac{-6 + 24}{\sqrt{9 + 16} \cdot \sqrt{4 + 36}} =$$

$$= \frac{18}{\sqrt{25} \cdot \sqrt{40}} = \frac{18}{5 \cdot 2\sqrt{10}} = \frac{9}{5\sqrt{10}} \Rightarrow \alpha = 55^\circ 18' 17,45''$$

b) $\vec{a} (1, 4, -2)$
 $\vec{b} (2, -2, -3)$

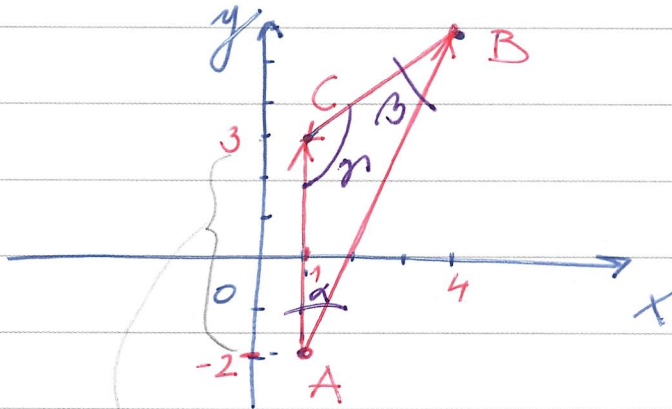


$$\cos(\vec{a}, \vec{b}) = \frac{1 \cdot 2 + 4 \cdot (-2) + (-2) \cdot (-3)}{\sqrt{1 + 16 + 4} \cdot \sqrt{4 + 4 + 9}} = \frac{2 - 8 + 6}{\sqrt{21} \cdot \sqrt{17}} = 0$$

$$\Rightarrow \alpha = 90^\circ$$

ALGORITHMY

(PRP) a) $A[1, -2]; B[4, 6]; C[1, 3]$



$$\vec{AC} = C - A = (1 - 1; 3 - (-2)) = (0, 5)$$

$$\vec{AB} = B - A = (4 - 1; 6 - (-2)) = (3, 8)$$

$$\vec{BC} = C - B = (1 - 4; 3 - 6) = (-3, -3)$$

$$\|\vec{AC}\| = \sqrt{0^2 + 5^2} = 5 \quad \text{SEMI TO AD PODLA OBR}$$

$$\|\vec{AB}\| = \sqrt{3^2 + 8^2} = \sqrt{9 + 64} = \sqrt{73}$$

$$\|\vec{BC}\| = \sqrt{(-3)^2 + (-3)^2} = \sqrt{9 + 9} = \sqrt{18} = \sqrt{9 \cdot 2} = 3\sqrt{2}$$

$$\cos \alpha = \frac{0 \cdot 3 + 5 \cdot 8}{5 \cdot \sqrt{73}} = \frac{40}{5\sqrt{73}} = \frac{8}{\sqrt{73}} \Rightarrow \alpha = 20^\circ 33' 21,76''$$

$$\cos \beta = \frac{3(-3) + 8(-3)}{\sqrt{73} \cdot 3\sqrt{2}} = \frac{-9 - 24}{3\sqrt{146}} = \frac{-33}{3\sqrt{146}} \Rightarrow$$

$$\Rightarrow \beta = 24^\circ 26' 38,24''$$

$$\cos \gamma = \frac{-3 \cdot 0 - 3 \cdot 5}{3\sqrt{2} \cdot 5} = \frac{-15}{15\sqrt{2}} = -\frac{1}{\sqrt{2}} \Rightarrow \gamma = 135^\circ$$

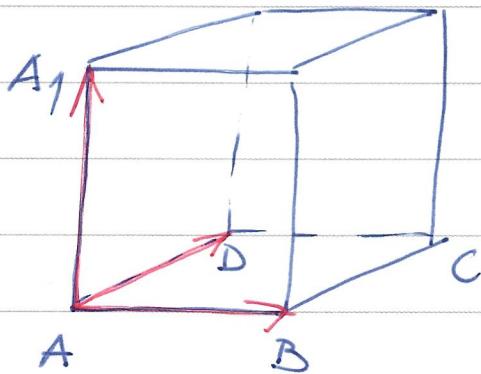
$$\text{SK: } \alpha + \beta + \gamma = 20^\circ 33' 21,76'' + 24^\circ 26' 38,24'' + 135^\circ = 180^\circ 0' 0''$$

ALGORYTMY

PR 9

$$A[1, -3, 4]; B[-2, 0, -1]; D[3, -1, 0]$$

$$A_1[0, 4, -2]$$



$$\begin{aligned}\vec{AB} &= B - A = (-2 - 1; 0 - (-3); -1 - 4) = \\ &= (-3, 3, -5)\end{aligned}$$

$$\begin{aligned}\vec{AD} &= D - A = (3 - 1; -1 - (-3); 0 - 4) = \\ &= (2, 2, -4)\end{aligned}$$

$$\begin{aligned}\vec{AA_1} &= (0 - 1; 4 - (-3); -2 - 4) = \\ &= (-1, 7, -6)\end{aligned}$$

$$V = |(\vec{AB} \times \vec{AD}) \cdot \vec{AA_1}| =$$

$$\vec{AB} \times \vec{AD} = \begin{vmatrix} -3 & 3 & -5 \\ 2 & 2 & -4 \end{vmatrix} =$$

$$= (-12 + 10; -10 - 12; -6 - 6) = (-2, -22, -12)$$

$$= |(-2, -22, -12) \cdot (-1, 7, -6)| = |2 - 154 + 72| =$$

$$= |-80| = 80 \text{ m}^3$$

⇒ OBJEM ROWNOBEEŽNO-STĚNA

ALGORITMY

$$S = 2 (\|\vec{AB} \times \vec{AD}\| + \|\vec{AB} \times \vec{AA_1}\| + \|\vec{AD} \times \vec{AA_1}\|)$$

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$$\vec{AB} \times \vec{AA_1} = \begin{vmatrix} \cancel{-3} & 3 & -5 & -3 & 3 & \cancel{-5} \\ \cancel{-1} & 7 & -6 & -1 & 7 & \cancel{-6} \end{vmatrix} =$$

$$= (-18 + 35, 5 - 18, -21 + 3) = \underline{\underline{(17, -13, -18)}}$$

$$\vec{AD} \times \vec{AA_1} = \begin{vmatrix} \cancel{2} & 2 & -4 & 2 & 2 & \cancel{-4} \\ \cancel{-1} & 7 & -6 & -1 & 7 & \cancel{-6} \end{vmatrix} =$$

$$= (-12 + 28, 4 + 12, 14 + 2) = \underline{\underline{(16, 16, 16)}}$$

$$S = 2 (\|(-4, -22, -12)\| + \|(17, -13, -18)\| + \|(16, 16, 16)\|)$$

$$= 2 (\sqrt{4 + 484 + 144} + \sqrt{289 + 169 + 324} + \sqrt{3 \cdot 256})$$

$$= 2 (\sqrt{632} + \sqrt{782} + 16\sqrt{3}) \div 2 \cdot 80,82 =$$

$$= \underline{\underline{161,63 \text{ m}^2}}$$

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ORSAH POVRCHU

ROVNOBEŽNOSTENA